

## 第二章 函数的导数(Derivative)

## 2.1 导数的定义(The Definition of Derivative)



## Multiple Choice Questions

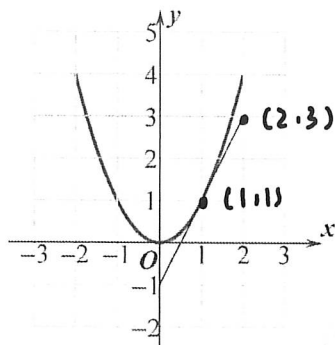
1. Which of the following values is the average rate of change of  $f(x) = \sqrt{x+2}$  over the interval  $[-1, 2]$ ?

- (A) -4                      (B)  $-\frac{1}{3}$                       (C)  $\frac{1}{3}$                       (D) 3

2. Let  $f(x) = 4 - 3x$ . Which of the following is equal to  $f'(0)$ ?

- (A) 4                      (B) -4                      (C) -3                      (D) 4

3. The graph of  $f$  and its tangent line at  $x=1$  is shown below, find  $f'(1) = (\quad)$ .



- (A) 0                      (B)  $\frac{1}{2}$                       (C) 1                      (D) 2

4. Let  $f(x) = kx^3 - 2x$ , ( $k \neq 0$ ). The average rate of  $f(x)$  over the interval  $[-2, -1]$  is 3, find  $k$ .

- (A)  $\frac{5}{7}$                       (B)  $\frac{1}{7}$                       (C)  $-\frac{1}{7}$                       (D)  $-\frac{5}{7}$

5. **Calculator** Let  $y = \tan e^x - x^2 + 1$ . Find  $\frac{dy}{dx} \big|_{x=0.3} =$

- (A) -12.056                      (B) 0                      (C) 271.066                      (D) 273.066

6. **Calculator** The graph of  $y = \sqrt{x^2+1} - 3x$  crosses the  $x$ -axis at one point in the interval  $[0, 1]$ . What is the slope of the graph at this point?

- (A) 0.354                      (B) -0.001                      (C) -2.666                      (D) 0

7. The population of insects at time  $t$  days is modeled by the function  $P$  with first derivative  $P'(t) = 0.1t^2 + 10t + 120$ . Which of the following statements is true?

- (A) At time  $t = 10$ , the population of insects is 230.  
 (B) At time  $t = 10$ , the population of insects is increasing at a rate of 230 insects per day.  
 (C) At time  $t = 10$ , the population of insects is decreasing at a rate of 230 insects per day.  
 (D) At time  $t = 10$ , the rate of change of the number of insects is increasing at a rate of 230 insects per day.

8. If the line tangent to the graph of a function  $g$  at the point  $(1, 4)$  passes through the point  $(-2, -5)$ , what is the slope of this line?

- (A) 3 (B) -3 (C)  $\frac{1}{3}$  (D)  $-\frac{1}{3}$



## Free Response Questions

9. Let  $f(x) = x^2 - 2x$

(a) Find the average rate of change of  $f$  over  $[1, 3]$ .

(b) Find  $\frac{f(2+h) - f(2)}{h}$ .

(c) Find the instantaneous rate of change of  $f$  at  $x = 2$ .

9. a.  $\frac{9-6-(1-2)}{3-1} = 2$       c.  $\lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h}$

b.  $\frac{(2+h)^2 - 2(2+h) - (4-4)}{h} = \lim_{h \rightarrow 0} \frac{h^2 + 4h + 4 - 4 - 4h}{h} = \lim_{h \rightarrow 0} \frac{h^2}{h} = 2$

$= h + 2$

10. Use the definition of derivative to find the derivative of  $f(x) = x^3$ , then find  $f'(2)$ .

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{(x+h)^3 - x^3}{h} \\ &= \lim_{h \rightarrow 0} \frac{h^3 + 3x^2h + 3xh^2 + h^3}{h} \\ &= 3x^2 \\ f'(2) &= 3 \times 2^2 = 12. \end{aligned}$$